

**TOP
SECRET**

SNAP CIRCUITS

LEARN. BUILD. PLAY.

SPY KIT

VIBRATION
SENSOR



MOTION
DETECTOR



TRIP
WIRE



VOICE
CHANGER



25+ SPY MISSIONS!



Snap Circuits® uses building blocks with snaps to assemble real electronic circuits. Each block has a function: switch, light, battery, different length wire blocks, and more! With easy-to-follow project manual, **simply snap together to create working circuits!**



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For Additional Projects



Go to:
www.elenco.com/scspy1-manuals/
to download projects 9-25

Download projects 9-25, plus more information, troubleshooting, about your parts and how circuits work.



Installing batteries:

- Remove the safety cover from the battery holder (if your holder has one) with a Phillips head screwdriver
- Insert three (3) 1.5V AA batteries (not included) in the holder. Be sure to orient the battery "+" as marked in the holder.
- Replace the cover (if your holder had one). When installing a battery, be sure the spring is compressed straight back, and not bent up, down, or to one side.
- Battery installation should be supervised by an adult.

What is electricity? Nobody really knows. We only know how to produce it, understand its properties, and how to control it. Electricity is a form of energy that can give things the ability to move and work. Everything in the world around us is made of particles called protons, neutrons and electrons. These three tiny particles are found in everything around us. When the electrons move, they create electricity.

Important Safety Notes for Parents and Kids

- **Adult Supervision:** Because children's abilities vary so much, even with age groups, adults should exercise discretion as to which experiments are suitable and safe. Make sure your child reads and follows all of the relevant instructions and safety procedures, and keeps them at hand for reference.
- This product is intended for use by adults and children who have attained sufficient maturity to read and follow directions and warnings.
- Save the packaging and instructions. They contain important information.
- **Warning: Choking Hazard**- Small parts. Not for children under 3 years of age.
- **Warning:** This product produces flashes that may trigger epilepsy in sensitized individuals.
- **Warning: Shock Hazard** - Never connect Snap Circuits® to the electrical outlets in your home in any way!
- **Warning:** Always check your wiring before turning on a circuit. Never leave a circuit unattended while the batteries are installed. Never connect additional batteries or any other power sources to your circuits.
- Never modify your parts, as doing so may disable important safety features in them, and could put your child at risk of injury
- Save the packaging and instructions. They contain important information.



Batteries: X3



- Use only 1.5V AA type, alkaline batteries (not included).
- Insert batteries with correct polarity.
- Do not mix old and new batteries.
- Remove batteries when they are used up.
- Do not mix alkaline, standard (carbon-zinc), or rechargeable (nickel-cadmium batteries).
- Do not connect batteries or battery holders in parallel.
- Do not short circuit the battery terminals.
- Never throw batteries in a fire or attempt to open its outer casing.
- Batteries are harmful if swallowed, so keep away from small children.
- Non-rechargeable batteries should not be recharged. Rechargeable batteries should only be charged under adult supervision, and should not be recharged while in the product.

Helpful Symbols:
Look for these symbols throughout the manual.



GENERAL INFORMATION
• Definitions
• "How it works?"



SNAP CIRCUITS
Technical information using your parts.



ADULT ASSISTANCE REQUIRED
Get an adult to help with your build

Let's get your secret agent training started!



Hello! I'm Dr. Spark and I'm excited to welcome you to the SNAP CIRCUITS SPY KIT! I'm the lead scientist working in the Spy Lab where I develop the amazing gadgets that make spying fun. My gadgets are designed for young spies-in-training. The Spy Kit is packed with features....."

SNAP CIRCUITS



TOP SECRET

SNAP CIRCUITS

Designed for young spies-in-training, the **SPY Kit** is packed with features that will ignite your imagination and sharpen your spy skills. Get ready to explore the secrets of espionage and become a master spy as you dive into hands-on projects and engaging activities. The **SPY Kit** is not just a toy—it's a gateway to discovering the fascinating world of spy gadgets and covert operations!



IMPORTANT:
If any parts are missing or damaged,
DO NOT RETURN TO RETAILER.

e-mail us at:
support@elenco.com
You may order additional/
replacement parts:
elenco.com/replacement-parts

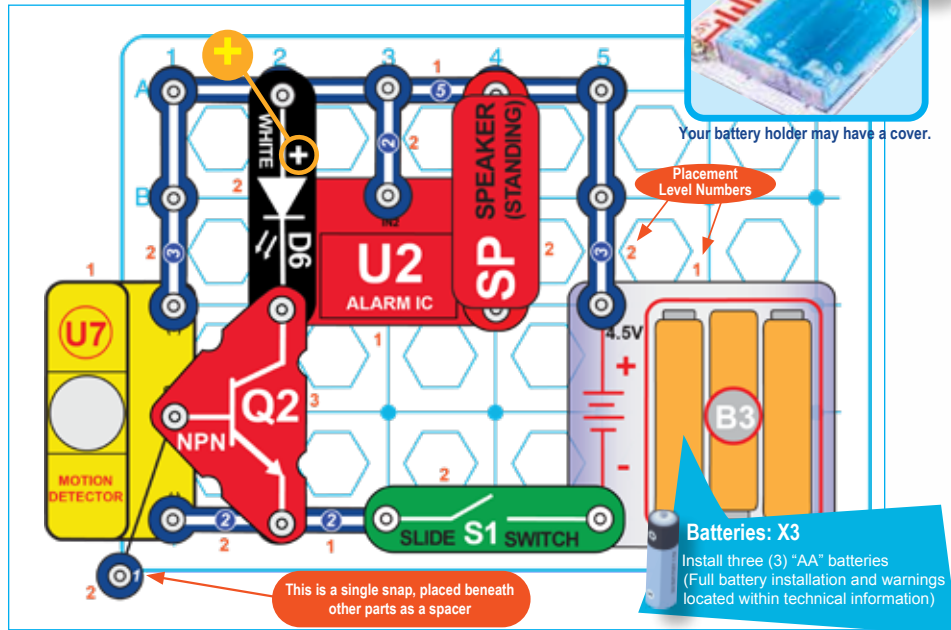
SPY KIT

Model SCSPY1

PARTS LIST

QTY.	ID	NAME	Part #
☐ 1		Base Grid (7.7x5.5)	6SCBGBK
☐ 1	1	1-snap wire	6SC01
☐ 3	2	2-snap wire	6SC02
☐ 3	3	3-snap wire	6SC03
☐ 1	5	5-snap wire	6SC05
☐ 1	B3 B3B	3AA Battery holder	6SCB3
☐ 1	C5	Capacitor 470 μ F	6SCC5
☐ 1	D6	White LED	6SCD6
☐ 1		Red/Green/Blue Filters	6SCFRGB
☐ 1		Jumper Wire Black	6SCJ1
☐ 1		Jumper Wire Red	6SCJ2
☐ 1	Q2	NPN Transistor	6SCQ2
☐ 1	Q3	SCR	6SCQ3
☐ 1	Q4	Phototransistor	6SCQ4
☐ 1	R3	Resistor 5.1k Ω	6SCR3
☐ 1	S1	Slide Switch	6SCS1
☐ 1	S2	Press Switch	6SCS2
☐ 1	S4	Vibration Switch	6SCS4
☐ 1	SP	Speaker	6SCSP
☐ 1	U2	Alarm IC	6SCU2
☐ 1	U7	Motion Detector IC	6SCU7
☐ 1	U34	Voice Changer IC	6SCU34
☐ 1	X1	Microphone	6SCX1
☐ 1		Blue Stand	626100

Mission 1: Motion Detector



Dr. Spark uses the motion detector to make sure there are no unauthorized people entering Secret Spy Kit lab. What will you use your motion detector for?

You could set up the motion detector in your room or play area to monitor movement. Test its sensitivity and range, using it to detect when someone enters your base. Experimenting with the motion detector will help you understand basic electronics and the principles of infrared technology.

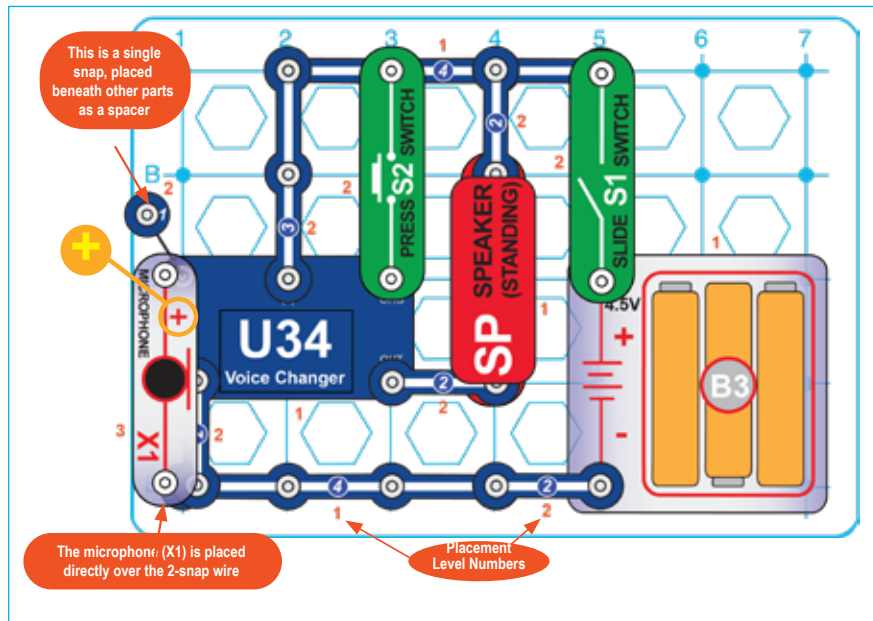
- Build the circuit shown by placing all the parts with a orange 1 next to them on the board first. Then, place parts marked with a 2, and then parts with a 3.
- Install three (3) "AA" batteries (not included) into the **Battery holder (B3 or B3B)** if you have not done so already. When installing a battery, be sure the spring is compressed straight back, and not bent up, down, or to one side.
- Then, assemble parts marked with a 2.
- Note that the **speaker (SP)** is standing up.
- Place the **base grid** into the blue stand (with the **slideswitch (S1)** touching the stand) and carefully stand near the edge of a table, facing across a room.
- Turn on the **Slide Switch (S1)**.
- An alarm sounds and a light shines for a few seconds on start-up, and then for 5-10 seconds whenever the circuit detects motion in the room.
- This circuit will work in the dark, but be careful not to hurt yourself moving around a room in the dark.

ADULT ASSISTANCE REQUIRED
Battery installation should be supervised by an adult.

The motion detector (U7) is designed to detect changes in infrared radiation, especially the type emitted by people.

Objects that generate heat, including people and animals, also produce infrared radiation. Infrared radiation cannot be seen with our eyes, but can be detected.

Mission 2: Voice Changer



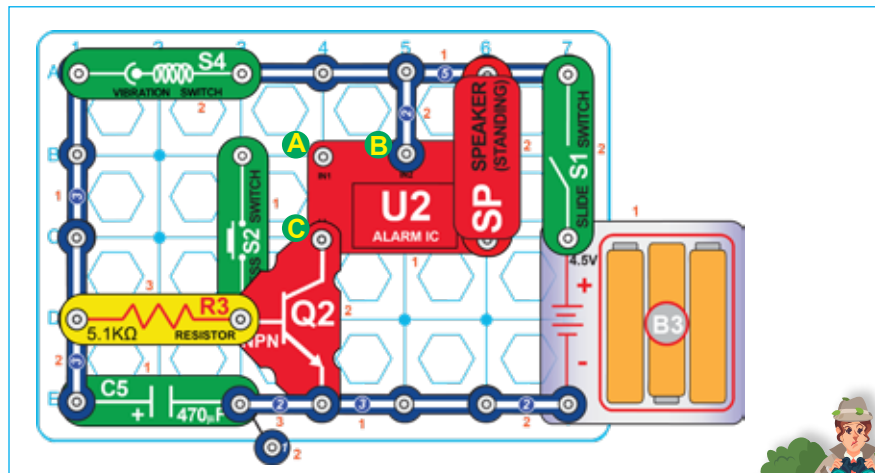
Dr. Spark knows that a fake voice can be vital to a successful mission.

Use the voice changer to play pranks, send secret messages, or role-play as their favorite spy characters.

- Build the circuit shown by placing all the parts with a orange 1 next to them on the board first. Then, place parts marked with a 2, and then parts with a 3. Note that a **2-snap wire** is underneath the **microphone (X1)**.
- Install three (3) "AA" batteries (not included) into the **Battery Holder (B3)** if you have not done so already.
- Turn on the **Slide Switch (S1)**.
- Push the **press switch (S2)** and talk near the **microphone (X1)**. Your voice gets played back at higher frequency.
- Press S2 a second time. Now your voice is played back at lower frequency.
- Press S2 a third time. Now your voice is played back with robotic tone.
- Press S2 a fourth time. The circuit beeps twice, signaling that it has been reset. Press again for higher frequency sound, twice for lower frequency, or three times for robotic sound.
- Eavesdropping: connect the microphone using the **red & black jumper wires**, then talk softly into the microphone to hear your voice on the speaker. If the jumper wires were much longer then you could use this to listen to conversations in another room or building.



Mission 3: Vibration Alarm



Dr. Spark suggests that you create alarms that notify yourself of any disturbances to protect your most valuable stuff. Kids can place the vibration sensor on different surfaces to see how it reacts to various movements. Enhancing their understanding of sensor technology and practical applications in security.

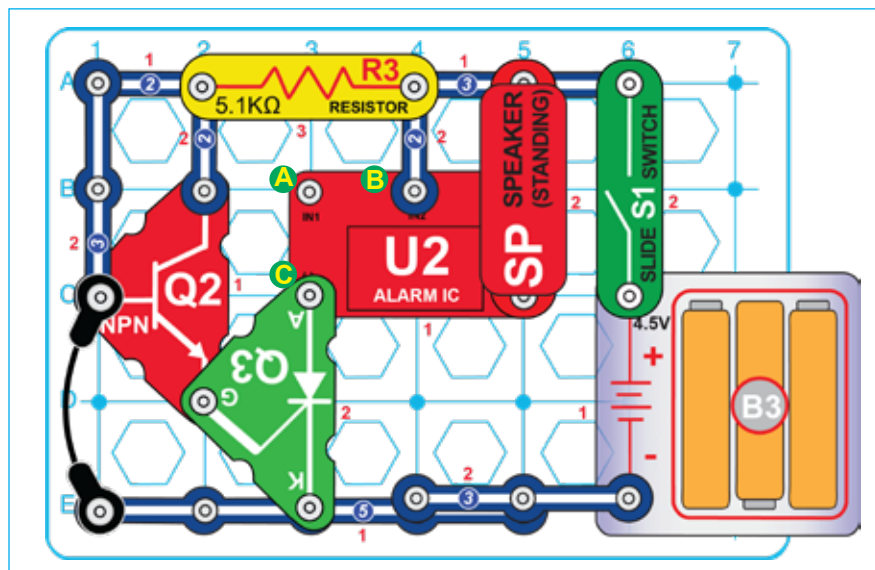
- Build the circuit as shown. Tap on the vibration **switch (S4)** or bang on the table to sound an alarm for a few seconds.
- Connect a **jumper wire** across points A & B or A & C to change the siren.
- Note: The **press switch (S2)** is used here as a 1-snap, so pressing it does nothing.



The vibration switch contains a spring and a contact switch. When shaken the spring bounces it briefly connects like a switch, which briefly connects the batteries to the 5.1kΩ resistor (R3) and the 470µF capacitor (C5). The resistor controls the alarm IC (U2) and speaker (SP) through the PNP transistor (Q2), producing sound.

Capacitors can store small amounts of electricity. The brief connection to the 470µF capacitor (C5) charges it, then it slowly discharges through the resistor and transistor, keeping the siren on for a few seconds after the vibration switch was shaken.

Mission 4: Trip Wire



Dr. Spark recommends that young spies set up trip wires around their secret lairs to catch intruders. A trip wire is crucial to alerting an agent of an unwanted trespasser or to protect your belongings. Experiment with different configurations and challenges to learn about tension and triggering mechanisms.

- Build the circuit as shown. Turn on the **slide switch (S1)**. Nothing happens. Disconnect the **black jumper wire** and a siren sounds. The siren stays on even if you reconnect the black jumper wire. Turn off the **slide switch** to shut off the siren.
- If you had a longer jumper wire then you could lay it across a doorway or room. If a burglar entered and tripped on the wire then a siren would sound. The siren would stay on even if the burglar reconnected the wire.
- Connect a **jumper wire** across points A & B or A & C to change the siren.

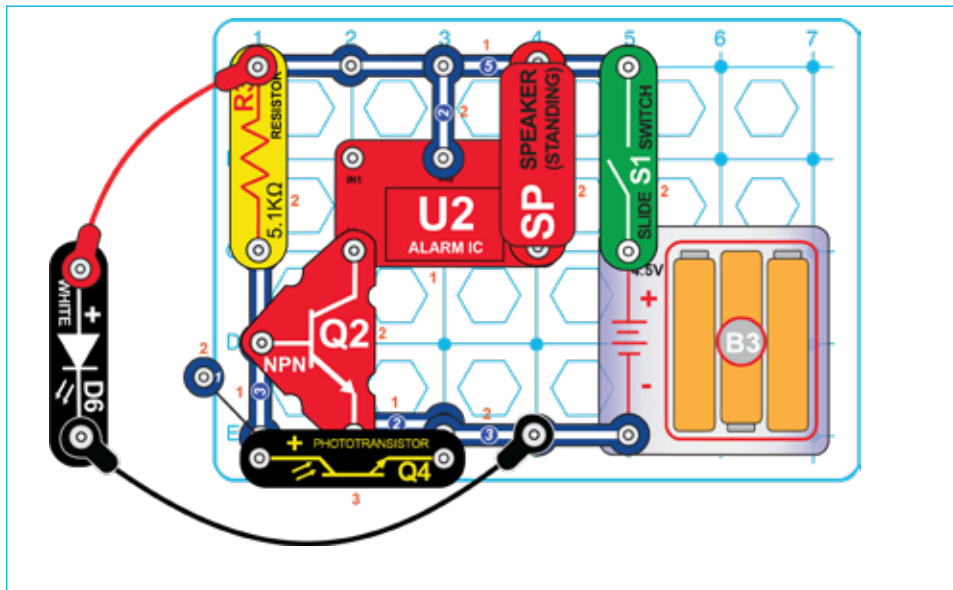


The SCR (Q3) is activated by a pulse at its G snap, then it stays on until the circuit is turned off. Here the SCR is connected to the alarm IC, so removing the black wire (the "trip" wire) activates the SCR and a siren. The SCR keeps the siren on even if the trip wire is reconnected.



This activity encourages problem-solving and creative thinking.

Mission 5: Break the Beam



Build the circuit shown.

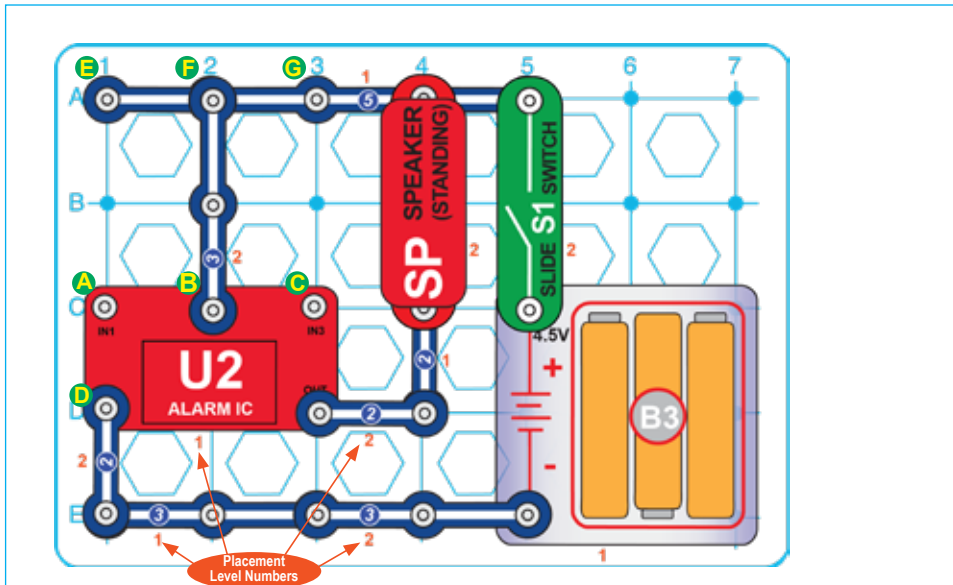
- Part A: Turn on the **slide switch (S1)**. The **white LED (D6)** is only on when the **press switch (S2)** is pushed.
- Part B: Replace the **press switch (S2)** with the **phototransistor (Q4, "+" on top)**. Cover and uncover Q4 to change the LED brightness. Compare the LED brightness in this circuit to part C of project 1, which did not have a transistor amplifying the current through Q4.
- Part C: In the original circuit, replace the **NPN transistor (Q2)** with the **SCR (Q3)**. Now the LED stays on after you release the press switch.



Circuits like this are used in many home security systems. A beam of light shines across a room to a sensor. An alarm sounds if an intruder enters the room and blocks the beam of light.



Mission 6: Sirens

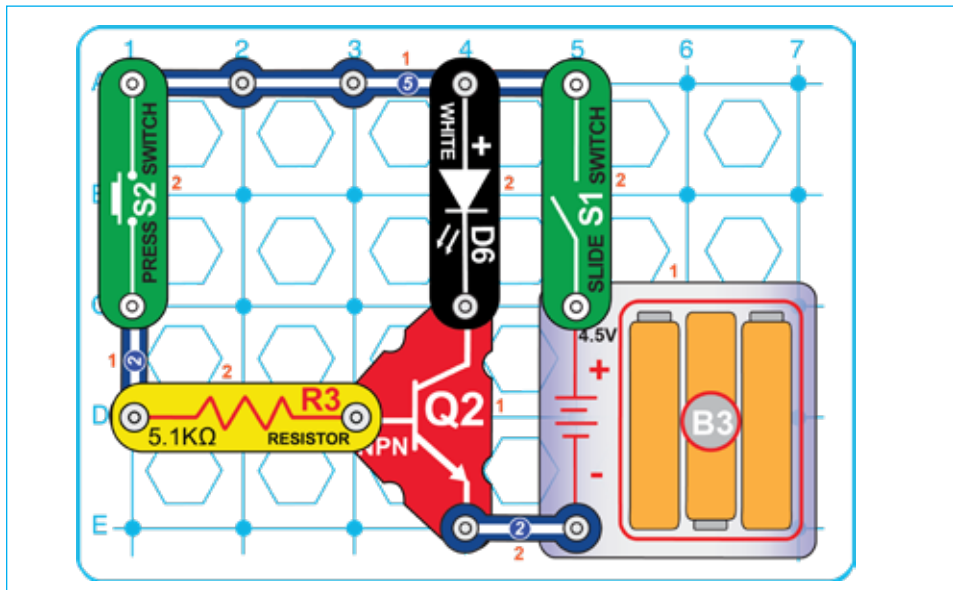


Build the circuit shown.

- Part A: Turn on the **slide switch (S1)**. A siren plays.
- Part B: Connect a jumper wire across snaps labeled A & B to make an ambulance sound, A & D to make a European siren, or B & C to make machine gun sound.
- Part C: With the original circuit, connect the **phototransistor (Q4)** across snaps E-A or G-C ("+" to E or G). Cover and uncover the phototransistor to change the sound.
- Part D: With the original circuit, replace the **3-snap wire** across snaps F-B with the phototransistor (Q4, "+" to F). A siren sounds when there is bright light on the phototransistor.
- Part E: With the original circuit, connect the **5.1kΩ resistor (R3)** across snaps G-C and replace the speaker with the white LED (D6, "+" on top). The LED flashes.



Mission 7: Transistor Amplifier



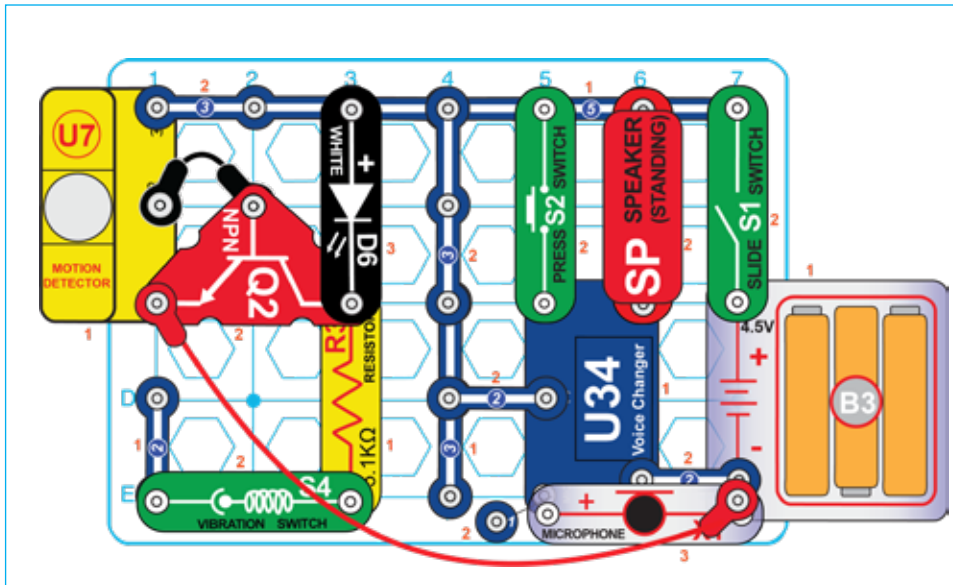
Turn on the **slide switch (S1)**. The **white LED (D6)** is only on when the **press switch (S2)** is pushed.

Part B: Replace the **press switch (S2)** with the **phototransistor (Q4, "+" on top)**. Cover and uncover Q4 to change the LED brightness. Compare the LED brightness in this circuit to part C of project 1, which did not have a transistor amplifying the current through Q4.

Part C: In the original circuit, replace the **NPN transistor (Q2)** with the **SCR (Q3)**. Now the LED stays on after you release the press switch.



Mission 8: Combo Circuit



Have fun and experiment with different settings to see how changing their voice can alter their disguise and add an element of surprise to their missions.

This circuit combines a motion detector and a voice changer. Build the circuit, place it into the blue stand, and carefully stand it up. Position it near the edge of a table, facing across a room.

Turn on the **slide switch (S1)**. The **white LED (D6)** lights for a few seconds on start-up, and then whenever the circuit detects motion in the room.

Voice Changer:

- Your voice gets played back at higher frequency.
- Press S2 a second time. Now your voice is played back at lower frequency.
- Press S2 a third time. Now your voice is played back with robotic tone.
- Press S2 a fourth time. The circuit beeps twice, signaling that it has been reset. Press again for higher frequency sound, twice for lower frequency, or three times for robotic sound.

The **vibration switch (S4)** is used here for stability, and is not electrically connected.

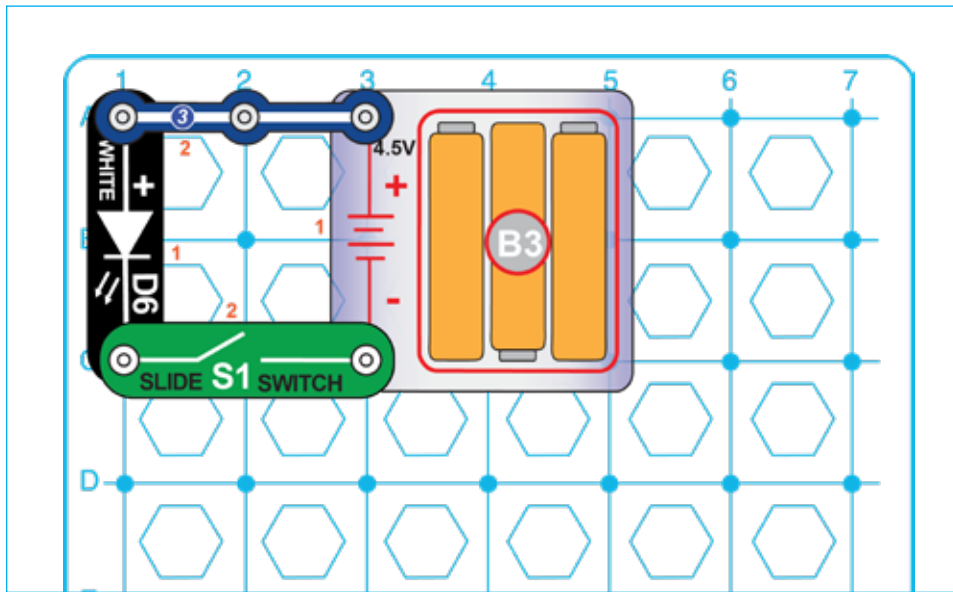


This circuit is shown on the box and manual cover, use that as a guide in building it



Go to:
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to download projects 9-25.

Project 9: Flashlight

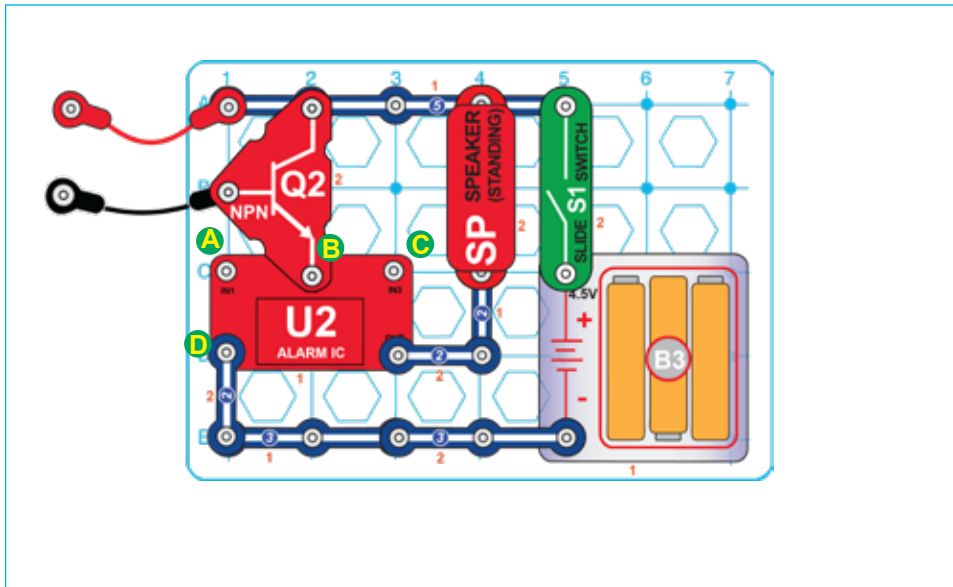


- Turn on the Slide Switch (S1), and the White LED (D6) lights.
- Part B, Vibration Flicker Light: Replace the slide switch (S1) with the vibration switch (S4). Tap on S4 or bang on the table to make the LED flash.
- Part C: Light Controlled Light: Replace S4 with the phototransistor (Q4, "+" on left). Vary the amount of light shining on Q4 to make the LED brighter or dimmer.



When you turn on the slide switch, electricity flows from the batteries, through the white LED and back to the battery through the switch. If the switch is off, the flow of electricity is blocked, and the white LED won't light.

Project 10: Water Alarm

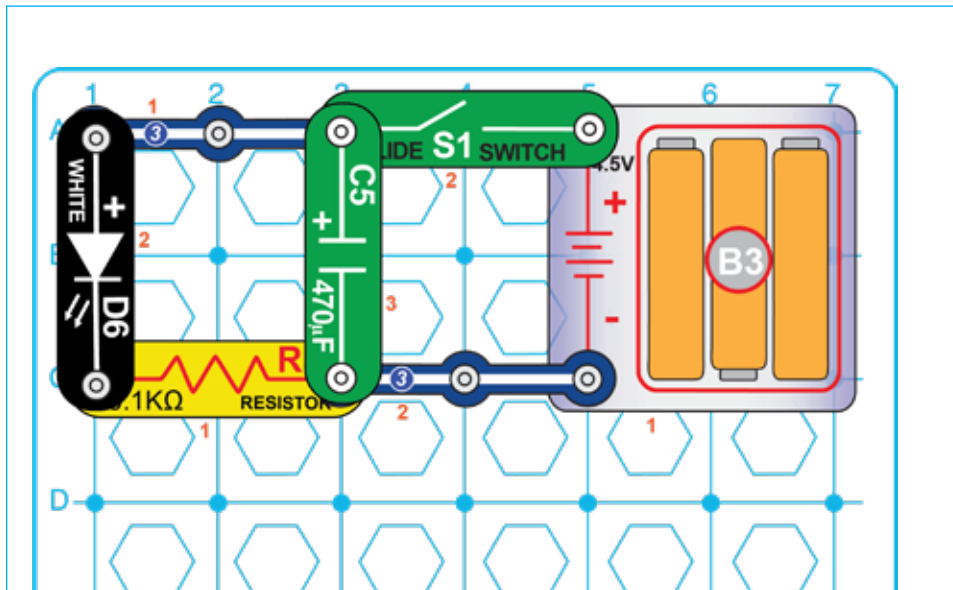


- Turn on the slide switch (S1). Place the loose ends of the jumper wires in water to play a siren. You can use a cup or water or a small puddle on the table.
- You can also activate the siren by touching the metal in the ends of the wires with your fingers, but you may need to wet your fingers.
- Connect a jumper wire across snaps labeled A & B to make ambulance sound, A & D to make European siren, or B & C to make machine gun sound.
- Don't drink any water used here.



Normal tap water has some electrical resistance but much less than air. Your body is mostly water, so has similar resistance to water.

Project 11: Mini Rechargeable Battery



Turn on the slide switch (S1). The white LED (D6) lights and the 470μF capacitor (C5) charges up.

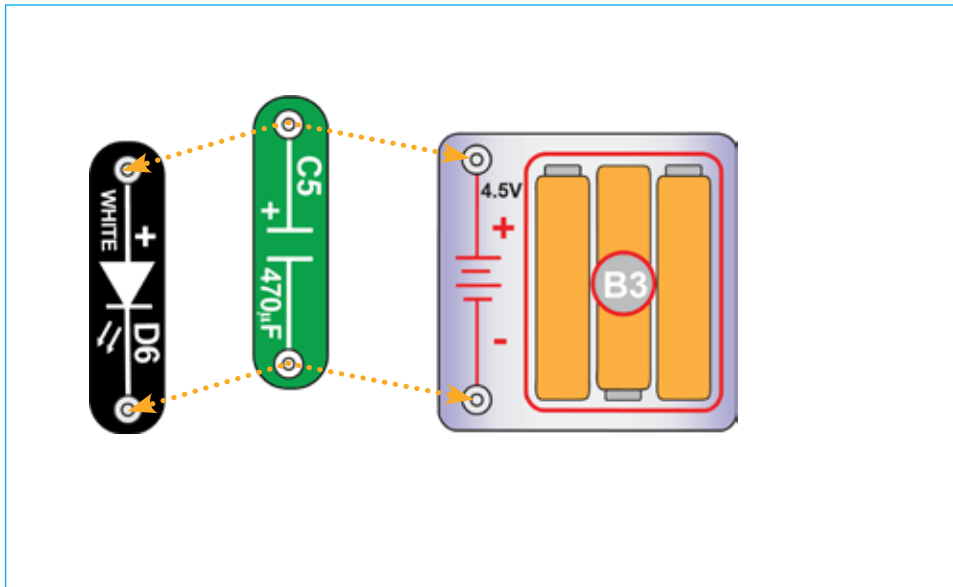
Turn off the switch, and notice how the LED stays on for a short time. Remove C5 and see how quickly the LED goes out when you turn off the switch.

If you replace the 5.1kΩ resistor (R3) with a 3-snap wire then the LED is brighter but shuts off faster.



The capacitor is like a mini rechargeable battery because it can store electricity. Capacitors store electricity in an electric field while batteries store it as chemical energy. Capacitors cannot store nearly as much electricity as rechargeable batteries, but they can be charged and discharged much faster.

Project 12: Mini Rechargeable Battery (II)



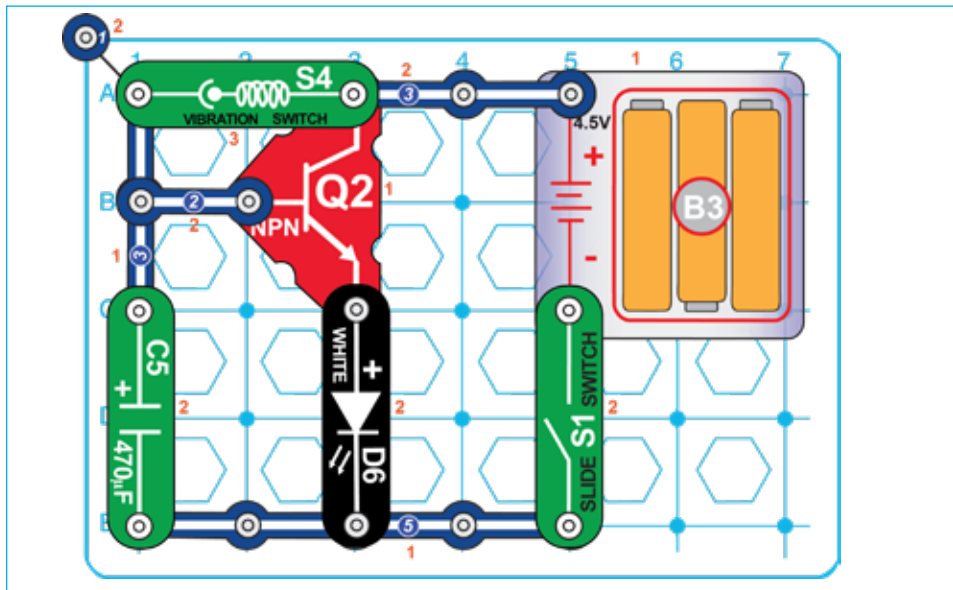
Touch the 470μF capacitor (C5) across the battery holder (B3) snaps, C5 "+" to B3 "+", to charge it. Next touch C5 across the white LED (D6) snaps, C5 "+" to D6 "+", to discharge it and light the LED. Repeat several times.



This circuit demonstrates that capacitors can store electricity. You charge up the capacitor using the batteries, then move the capacitor to the white LED, lighting it.



Project 13: Vibration Light

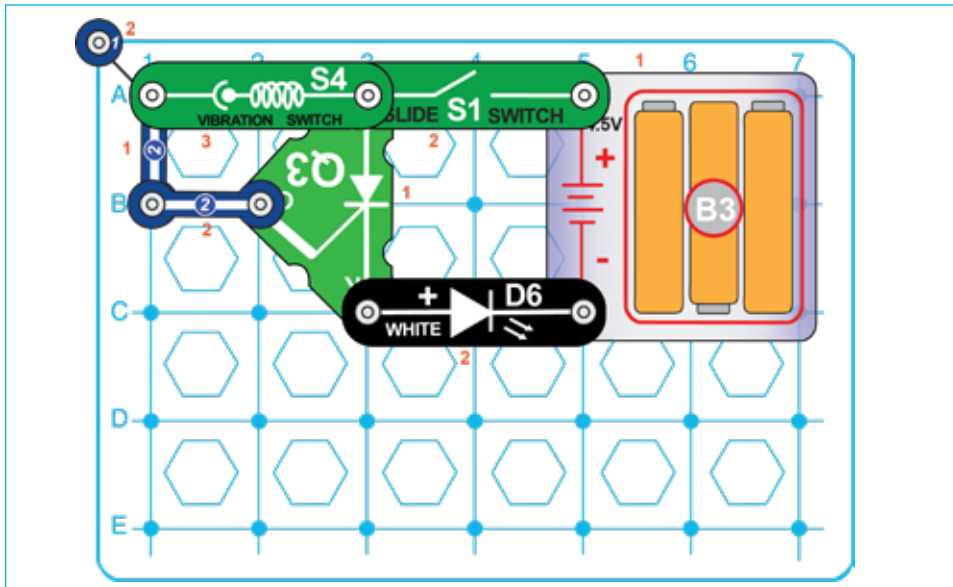


Turn on the slide switch (S1), then tap on the vibration switch (S4) or bang on the table to light the white LED (D6) for a few seconds.



When the vibration switch is activated it turns on the white LED and charges the 470µF capacitor (C5). The capacitor keeps the LED on for a few seconds after the vibration ends. If you remove C5 then the LED will only flicker for a moment.

Project 14: Vibration Light that Stays On



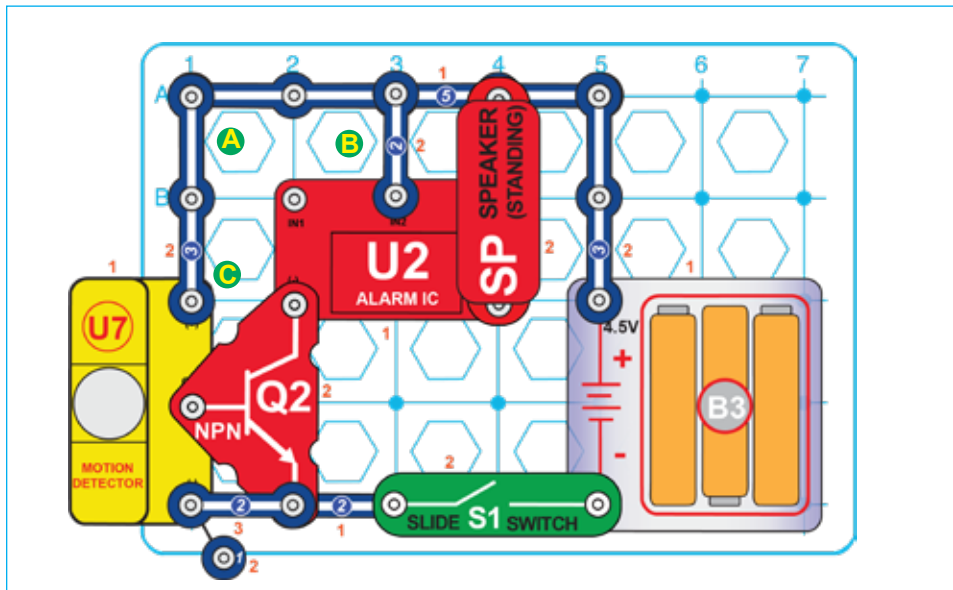
Turn on the slide switch (S1), then tap on the vibration switch (S4) or bang on the table to turn on the white LED (D6). The LED stays on until you turn off the slide switch.



When the vibration switch is activated it turns on the SCR (W3) and white LED. Once turned on the SCR stays on, keeping the LED on after the vibration ends.



Project 15: Motion Detector - Siren



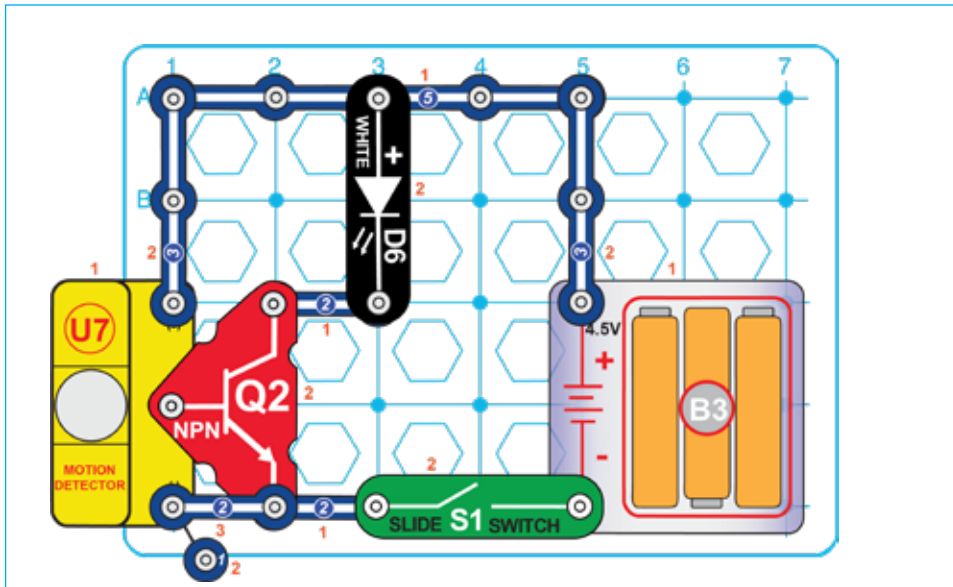
- Build the circuit, place the base grid into the blue stand (with the slide switch (S1) touching the stand), and carefully stand it up. Position it near the edge of a table, facing across a room.
- Turn on the **Slide Switch (S1)**.
- An alarm sounds for a few seconds on start-up, and then for 5-10 seconds whenever the circuit detects motion in the room.
- To change the siren sound, connect the red or black jumper wire between the snaps labeled A & B or A & C.
- This circuit will work in the dark, but be careful not to hurt yourself moving around a room in the dark.



The motion detector (U7) detects infrared radiation emitted by the heat in our bodies. A small output signal from the motion detector controls the alarm IC (U2) using the NPN transistor (Q2).

Mount circuit on the blue stand and face across a room.

Project 16: Motion Detector - Light

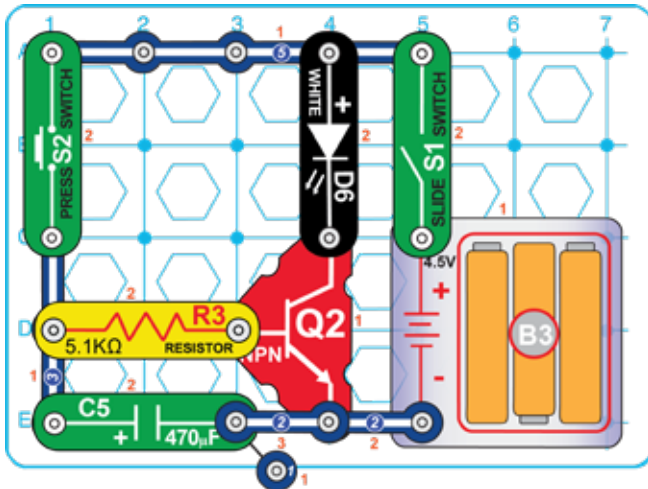


Build the circuit, place the base grid into the blue stand, and carefully stand it up. Position it near the edge of a table, facing across a room.

Turn on the slide switch (S1). The white LED (D6) lights for a few seconds on start-up, and then whenever the circuit detects motion in the room.

Mount circuit on the blue stand and face across a room.

Project 17: Time Delay Light



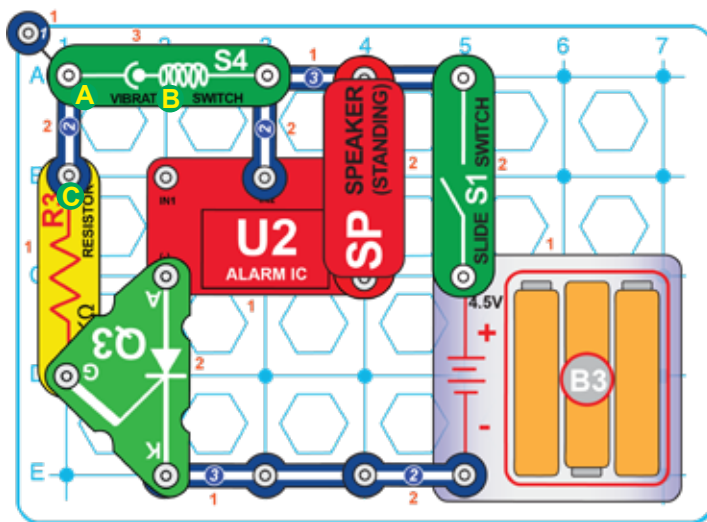
Turn on the slide switch (S1), then press and release the press switch (S2). The white LED (D6) turns on and slowly turns off.



Pressing S2 charges the 470μF capacitor (C5). With S2 released, C5 slowly discharges through the 5.1kΩ resistor (R3) and NPN transistor (Q2). The transistor controls the white LED, keeping it on while the capacitor is discharging.



Project 18: Vibration Alarm that Stays On

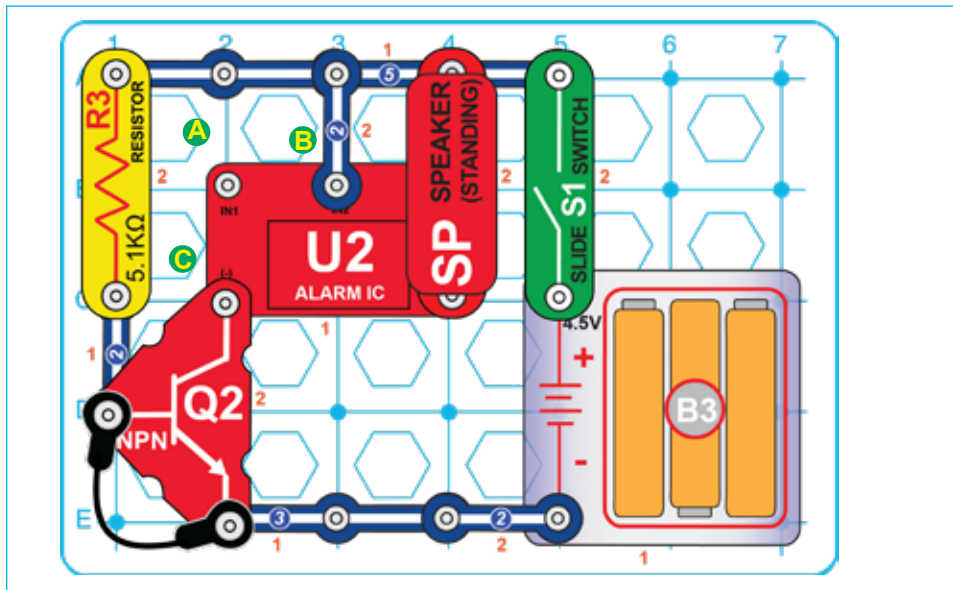


Turn on the slide switch (S1), then tap on the vibration switch (S4) or bang on the table to sound a siren. The siren stays on until you turn off the slide switch (S1). Connect a jumper wire across points A & B or A & C to change the siren.



When the vibration switch is activated it turns on the SCR (W3) and alarm IC (U2). Once turned on the SCR stays on, keeping the siren on after the vibration ends.

Project 19: Trip Wire Alarm that Shuts Off

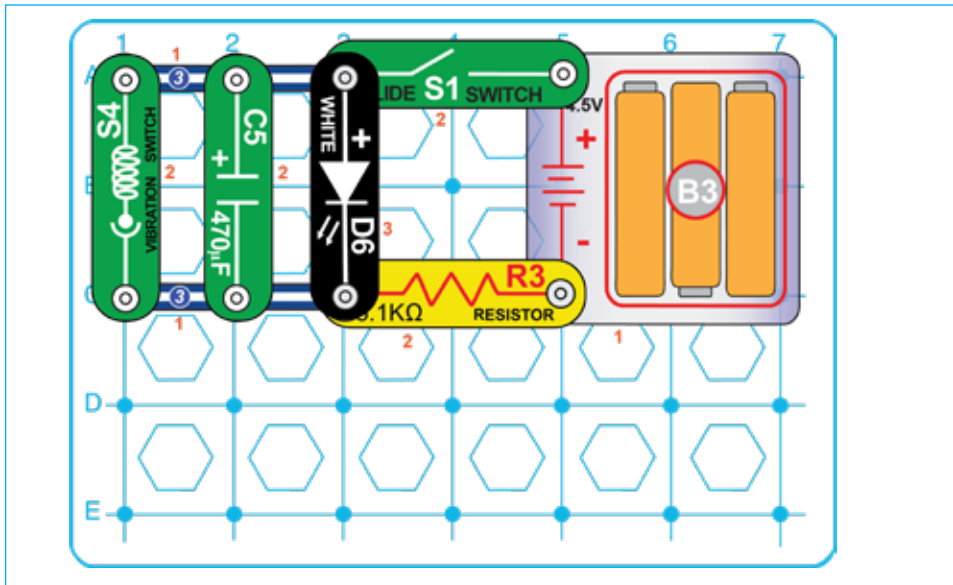


- Turn on the slide switch (S1). Nothing happens. Disconnect the black jumper wire and a siren sounds. Reconnect the black jumper wire to shut off the siren.
- If you had a longer jumper wire then you could lay it across a doorway or room. If a burglar entered and tripped on the wire then a siren would sound.
- Connect a jumper wire across points A & B or A & C to change the siren.



This circuit is similar to project 4, but the siren in project 4 stays on even if the black wire is reconnected.

Project 20: Vib Off

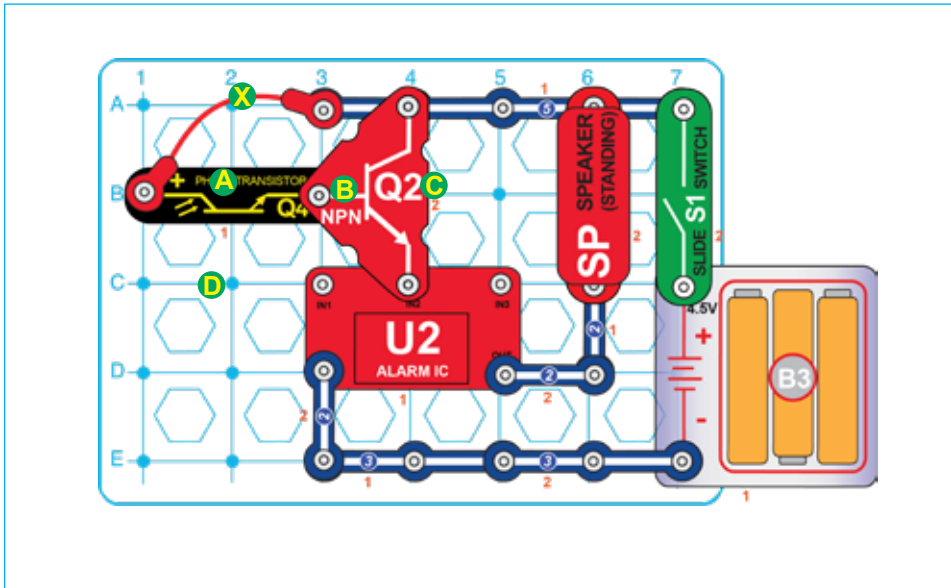


Build the circuit as shown and turn on the slide switch (S1). The white LED (D6) should be on. Tap on the vibration switch (S4) or bang on the table to make the LED blink (indicating that it is turning off briefly).



The vibration switch (S4) is connected so it bypasses the white LED (D6), so whenever vibrations turn on the switch, the LED turns off. However the vibrations occur so fast, turning off the LED so briefly that normally your eyes wouldn't even notice. The 470μF capacitor (C5) is used here to slow things down, so the LED does not immediately come back on after the vibration switch turned it off. If you remove C5, then you probably won't notice vibrations shutting off the LED (try it).

Project 21: Sensitive Light Controlled Siren



- Turn on the slide switch (S1). A siren plays if there is any light on the phototransistor (Q4).
- If the siren turns on too easily then add the 5.1kΩ resistor (R3) across between snaps labeled X & D to make it less sensitive to light.
- Connect a jumper wire across snaps labeled A & B to make ambulance sound, A & D to make European siren, or B & C to make machine gun sound.

Project 22: Secret Messages

For many years, nations have mostly been using fossil fuels, like coal, oil, and natural gas, as the way energy is used to make your life better. Energy lights your home at night. Energy heats your home in winter and cools it in summer. Energy runs your TV, computer, phone, and car. Energy keeps the food in your refrigerator fresh, heats your food in the microwave, or on the stove and cleans the dishes in your dishwasher. Energy powers these sources of the manufacturing of just about everything in your home, and in transporting it to your home. Change or all pollution problems and cannot be used up.

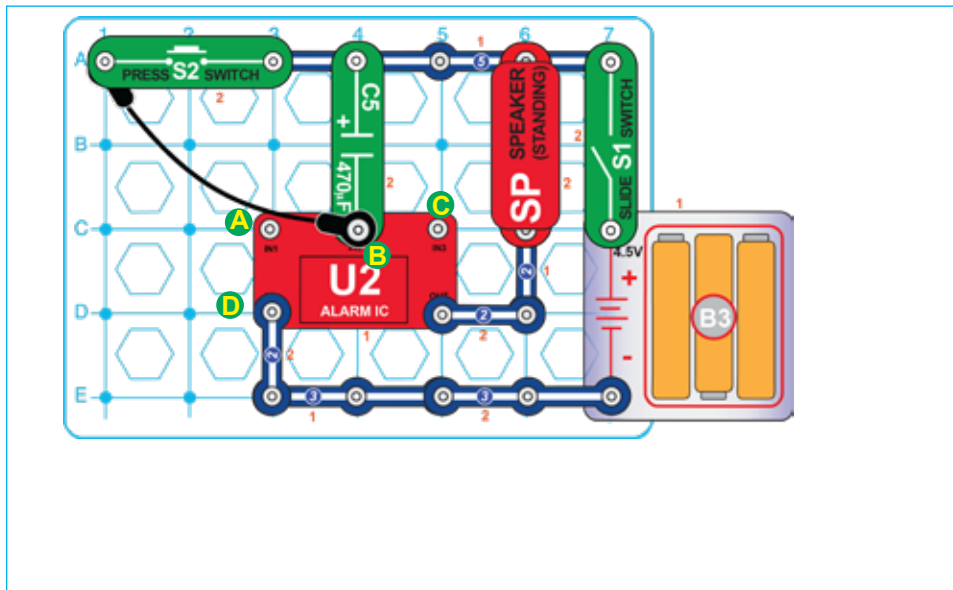
Everyone in the government has to do things. Engineers need to consider the long-term impact of our energy sources and work together in the climate on how to possibly use clean, renewable sources of energy. Everyone has to pay a price. Reducing their energy use. Government can help fund research on new technologies and encourage people to use them. And there are 565 million people in the world. A hundred billion dollars a year can't buy enough energy to do everything. Various components in a circuit are arranged in series and parallel configurations to control the electricity to do what we want. Circuits can be as simple as turning on a light bulb or as complex as running a computer or television.

Try reading these messages by looking through the red, green, or blue filters.
Create your own secret messages using crayons or markers.

The name electricity came from the Greek name for amber, the material in which electrical effects were first observed. What do you think of low voltage as being? Electricity is one of the fundamental forces of nature. At its most basic level, it is an attraction and repulsion between sub-atomic (very, very, very, very tiny) particles within a material. This attraction/repulsion is referred to as an electrical charge. It is similar to, and closely related to magnetism. These attractions/repulsions are extremely powerful but are so well balanced out at the sub-atomic

Energy cannot be created or destroyed, but can change into different forms. Kinetic energy is the energy of something in motion, like a moving car. Potential energy is energy waiting to make something happen, like water stored in a dam. Chemical energy is energy that can be harnessed from a fuel by chemical reaction, like gasoline or food, or batteries. Nuclear energy is energy that can be harnessed by splitting or combining atoms in a nuclear reaction. Heat, light, sound, and electricity are forms of kinetic energy, because the tiny particles they are made of are moving or vibrating to some degree.

Project 23: Short Siren



Turn on the slide switch (S1), then press and release the press switch (S2). A siren plays for a few seconds. Push S2 again to re-start the siren.

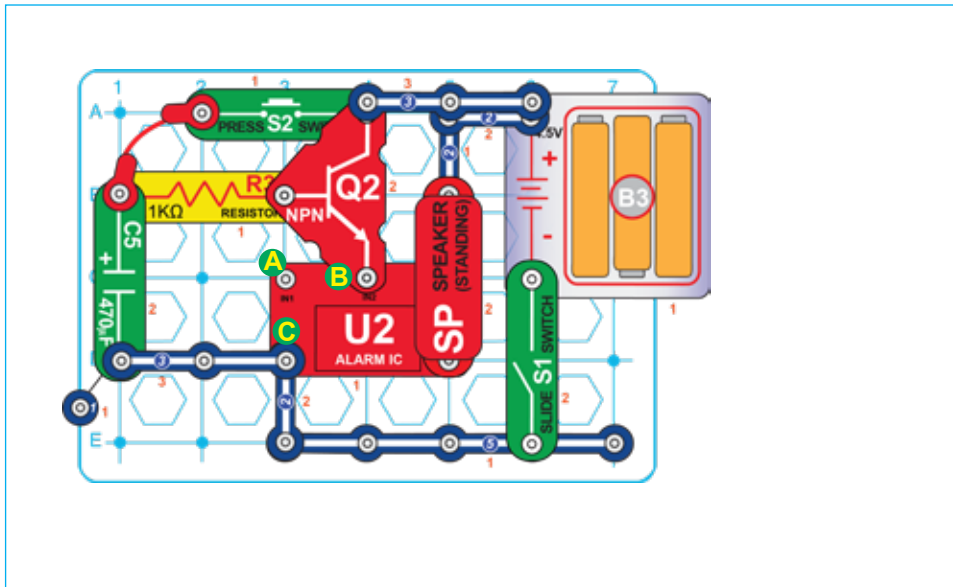
Connect the red jumper wire across snaps labeled A & B, A & D, or B & C to change the sound.



A siren plays for a few seconds while the 470µF capacitor (C5) charges up. Pressing S2 discharges the capacitor and restarts the siren.



Project 24: Long Siren



Turn on the slide switch (S1), then press and release the press switch (S2). A siren plays for several minutes. Push S2 again to re-start the siren.

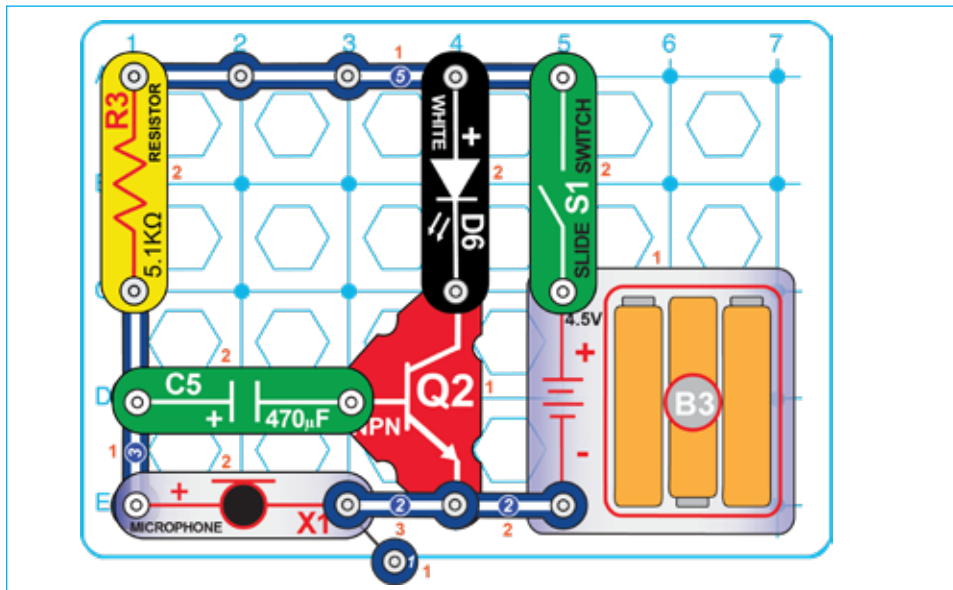
Connect the red jumper wire across snaps labeled A & B or A & C to change the sound.

Replace the speaker (SP) with the white LED (D6) to have a long light instead of a siren.



Pressing S2 charges the 470µF capacitor (C5), which then slowly discharges through the 1kΩ resistor (R3) and NPN transistor (Q2). The siren slowly fades away as the capacitor discharges.

Project 25: Blow the Light On and Off

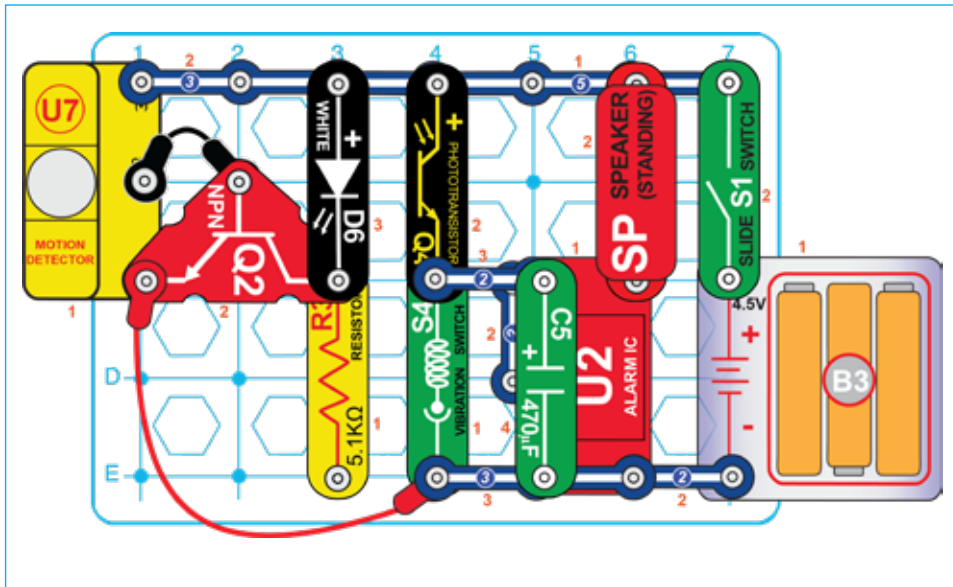


- Build the circuit and turn on the slide switch (S1).
- The white LED (D6) should be on, but slowly gets dimmer as the 470μF capacitor (C5) charges up.
- Blow directly into the microphone (X1) to change the LED brightness for a moment. The LED may get brighter or dimmer, depending on the the capacitor's current charge level.



Electric current flows through the 5.1kΩ resistor (R3) to charge up the 470μF capacitor (C5), and to the microphone (X1). The current through the capacitor also flows into the NPN transistor (Q2), which controls the white LED. The microphone is sensitive to sound waves, which are variations in air pressure, so blowing on the microphone changes its electrical resistance, which changes the current to the capacitor and transistor.

Project 26: Combo Circuit 2



- Build the circuit and turn on the slide switch (S1).
- The white LED (D6) lights for a few seconds on start-up, and then whenever the circuit detects motion in the room.
- A machine gun sound plays. Shine bright light on the phototransistor (Q4) to make the sound louder. Tap on the vibration switch (S4) to make the sound stop for a moment.

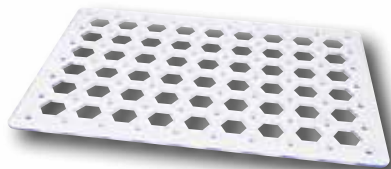


About Your Snap Circuits® Parts

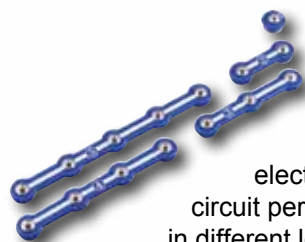
(Part designs are subject to change without notice).

BASE GRID

The **base grid** is a platform for mounting parts and wires. It functions like the printed circuit boards used in most electronic products, or like how the walls are used for mounting the electrical wiring in your home.



SNAP WIRES & JUMPER WIRES



The blue **snap wires** are wires used to connect components. They are used to transport electricity and do not affect circuit performance. They come in different lengths to allow orderly arrangement of connections on the base grid.

The **jumper wires** (red, black, orange, & blue) make flexible connections for times when using the snap wires would be difficult. They also are used to make connections off the base grid.



Wires transport electricity just like pipes are used to transport water. The colorful plastic coating protects them and prevents electricity from getting in or out.

BATTERY HOLDER

The **batteries (B3 or B3B)** produce an electrical voltage using a chemical reaction. This "voltage" can be thought of as electrical pressure, pushing electricity through a circuit just like a pump pushes water through pipes. This voltage is much lower and much safer than that used in your house wiring. Using more batteries increases the "pressure", therefore, more electricity flows.



B3B holders have a cover



Battery Holder (B3 or B3B)

SPEAKER

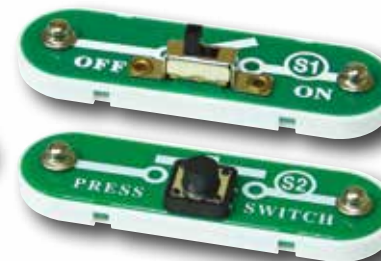
The **speaker (SP)** converts electricity into sound by making mechanical vibrations. These vibrations create variations in air pressure, which travel across the room. You "hear" sound when your ears feel these air pressure variations.



Speaker (SP)

SWITCHES

The **slide & press switches (S1 & S2)** connect (pressed or "ON") or disconnect (not pressed or "OFF") the wires in a circuit. When ON they have no effect on circuit performance. Switches turn on electricity just like a faucet turns on water from a pipe.



Slide & Press Switches (S1 & S2)

One side of the **vibration switch (S4)** connects to a spring, the other side connects to a wire through the spring. When the spring is shaken, the spring bounces to connect or disconnect the circuit.



Vibration Switch (S4)

RESISTOR

The **5.1kΩ resistor (R3)** "resists" the flow of electricity and is used to control or limit the current in a circuit. Materials like metal have very low resistance ($<1\Omega$), while materials like paper, plastic, and air have near-infinite resistance. Increasing circuit resistance reduces the flow of electricity.



About Your Snap Circuits® Parts

MICROPHONE

The **microphone (X1)** is actually a resistor that changes in value when changes in air pressure (sounds) apply pressure to its surface. Its resistance typically varies between $1k\Omega$ and $10k\Omega$.



Microphone (X1)

LED

The **white LED (D6)** is a light emitting diode, and may be thought of as a special one-way light bulb. In the "forward" direction, (indicated by the "arrow" in the symbol) electricity flows if the voltage exceeds a turn-on threshold brightness then increases. A high current will burn out an LED, so the current must be limited by other components in the circuit (Snap Circuits® LEDs have internal resistors added, to protect them in case you make wiring mistakes). LEDs block electricity in the "reverse" direction.



White LED (D6)

TRANSISTORS

The **NPN transistor (Q2)** uses a small electric current to control a large current, and is used in switching, amplifier, and buffering applications. Transistors are easy to miniaturize, and are the main building blocks of integrated circuits including the microprocessor and memory circuits in computers. **A high current may damage this part, so the current must be limited by other components in the circuit.**



NPN Transistor (Q2)

The **phototransistor (Q4)** is a transistor that uses light to control electric current.



Phototransistor (Q4)

SCR (Q3)

A - Anode
K - Cathode
G - Gate



The **SCR (Q3)** is a three-pin (anode, cathode and gate) silicon-controlled rectifier diode. Like other diodes, it permits current flow in only one direction. It will only conduct in the forward direction when triggered by a short pulse (or steady voltage applied) between the gate and cathode terminals. **A high current may damage this part, so the current must be limited by other components in the circuit.**

CAPACITOR

The **470 μ F capacitor (C5)** can store electrical pressure (voltage) for periods of time. This storage ability allows it to block stable voltage signals and pass changing ones. Capacitors are used for filtering and delay circuits.



Capacitor (C5)

About Your Snap Circuits® Parts

ELECTRONIC MODULES

The **alarm IC (U2)** and **voice changer (U34)** contain a specialized sound-generation integrated circuit (IC) and other supporting components (resistors, capacitors, and transistors) that are always needed with it.



Alarm IC (U2)

Connections:

IN1, IN2, IN3 - control inputs
(-) - power return to batteries
OUT - output connection

Connect control inputs to (+) power to make four alarm sounds (siren - changing in frequency (pitch), ambulance - rising and falling in frequency (pitch), European siren - alternating between two frequencies (pitches), machine gun - sound shuts off between bursts. See project 6 for an example of proper connections.

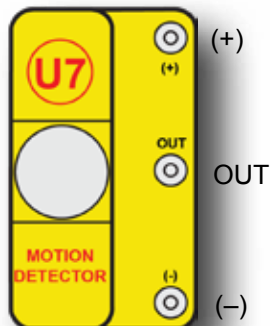


Voice Changer (U34)

Connections:

(+) - power from batteries
(-) - power return to batteries
MIC - microphone
CHG - change connection
OUT - output connection

The voice changer records voices and plays them back faster, slower, or with robotic tone. See project 2 for an example of proper connections and how to use it.



Motion Detector (U7)

Connections:

(+) - power from batteries
(-) - power return to batteries
OUT - output connection

The **motion detector (U7)** contains an infrared detector, amplifier-filter circuit, and timing circuit. See project 1 for an example of proper connections and how to use it.

All objects (including people and animals) produce infrared radiation due to the heat in them. Infrared radiation is similar to visible light but has a longer wavelength that our eyes cannot detect. The lens on top of the motion detector module filters and focuses the radiation, it is most sensitive to the radiation produced by our bodies.

Inside the motion detector module is an infrared detector with pyroelectric crystals, which create a tiny voltage when exposed to infrared radiation. A circuit amplifies and filters this voltage, but only responds to changes in the radiation level - so is only triggered by moving objects (motion). When motion is detected a timing circuit is used to control other snap circuits devices for a few seconds, such as an alarm.

Introduction to Electricity

What is electricity? Nobody really knows. We only know how to produce it, understand its properties, and how to control it. Electricity is the movement of sub-atomic charged particles (called **electrons**) through a material due to electrical pressure across the material, such as from a battery.

Power sources, such as batteries, push electricity through a circuit, like a pump pushes water through pipes. Wires carry electricity, like pipes carry water. Devices like LEDs, motors, and speakers use the energy in electricity to do things. Switches and transistors control the flow of electricity like valves and faucets control water. Resistors limit the flow of electricity.

The electrical pressure exerted by a battery or other power source is called **voltage** and is measured in **volts** (V). Notice the “+” and “-” signs on the battery; these indicate which direction the battery will “pump” the electricity.

The **electric current** is a measure of how fast electricity is flowing in a wire, just as the water current describes how fast water is flowing in a pipe. It is expressed in **amperes** (A) or **milliamps** (mA, 1/1000 of an ampere).

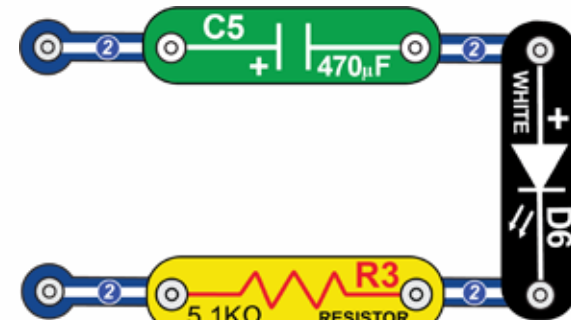
The “**power**” of electricity is a measure of how fast energy is moving through a wire. It is a combination of the voltage and current (Power = Voltage x Current). It is expressed in **watts** (W).

The **resistance** of a component or circuit represents how much it resists the electrical pressure (voltage) and limits the flow of electric current. The relationship is Voltage = Current x Resistance. When the resistance increases, less current flows. Resistance is measured in **ohms** (Ω), or **kilo ohms** (k Ω , 1000 ohms).

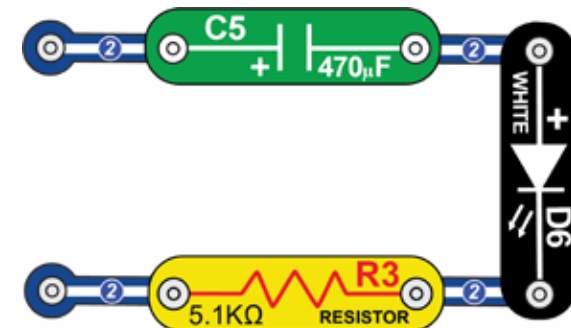
Nearly all of the electricity used in our world is produced at enormous generators driven by steam or water pressure. Wires are used to efficiently transport this energy to homes and businesses where it is used. Motors convert the electricity back into mechanical form to drive machinery and appliances. The most important aspect of electricity in our society is that it allows energy to be easily transported over distances.

Note that “distances” includes not just large distances but also tiny distances. Try to imagine a plumbing structure of the same complexity as the circuitry inside a portable radio - it would have to be large because we can't make water pipes so small. Electricity allows complex designs to be made very small.

There are two ways of arranging parts in a circuit, in series or in parallel. Here are examples:



Series Circuit



Parallel Circuit

Placing components in series increases the resistance; highest value dominates. Placing components in parallel decreases the resistance; lowest value dominates.

The parts within these series and parallel sub-circuits may be arranged in different ways without changing what the circuit does. Large circuits are made of combinations of smaller series and parallel circuits.

DOs and DON'Ts of Building Circuits

After building the circuits given in this booklet, you may wish to experiment on your own. Use the projects in this booklet as a guide, as many important design concepts are introduced throughout them. Every circuit will include a power source (the batteries), a resistance (which might be a speaker, LED (which has an internal protection resistor), integrated circuit, etc.), and wiring paths between them and back. **You must be careful not to create "short circuits" (very low-resistance paths across the batteries, see examples below) as this will damage components and/or quickly drain your batteries.** Only connect the ICs using configurations given in the projects, incorrectly doing so may damage them. **Elenco® is not responsible for parts damaged due to incorrect wiring.**

Here are some important guidelines:

ALWAYS USE EYE PROTECTION WHEN EXPERIMENTING ON YOUR OWN.

ALWAYS include at least one component that will limit the current through a circuit, such as the speaker, an LED (which has an internal protection resistor), or ICs (which must be connected properly).

ALWAYS use switches in conjunction with other components that will limit the current through them. Failure to do so will create a short circuit and/or damage those parts.

ALWAYS disconnect your batteries immediately and check your wiring if something appears to be getting hot.

ALWAYS check your wiring before turning on a circuit.

ALWAYS connect ICs using configurations given in the projects or as per the connection descriptions for the parts.

NEVER connect to an electrical outlet in your home in any way.

NEVER leave a circuit unattended when it is turned on.

For all of the projects given in this book, the parts may be arranged in different ways without changing the circuit. For example, the order of parts connected in series or in parallel does not matter — what matters is how combinations of these sub-circuits are arranged together.



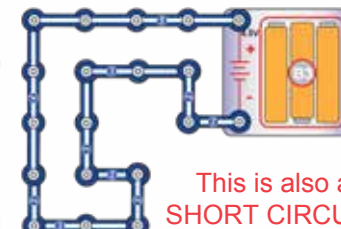
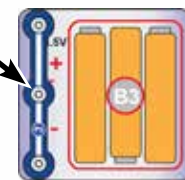
Warning to Snap Circuits® owners: Do not connect additional voltage sources from other sets, or you may damage your parts. Contact ELENCO® if you have questions or need guidance.

Examples of SHORT CIRCUITS - NEVER DO THESE!!!

Placing a 3-snap wire directly across the batteries is a **SHORT CIRCUIT**.



NEVER DO!



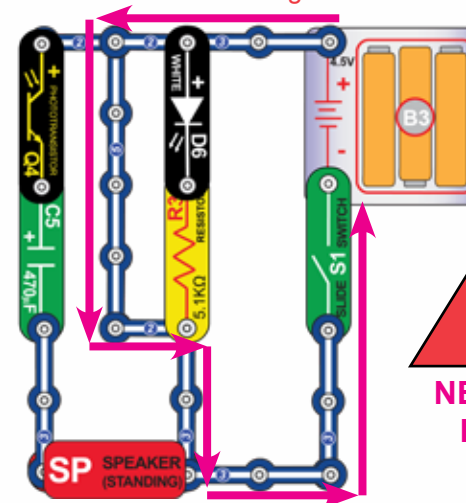
NEVER DO!

This is also a **SHORT CIRCUIT**.

When the slide switch (S1) is turned on, this large circuit has a **SHORT CIRCUIT** path (as shown by the arrows). The short circuit prevents any other portions of the circuit from ever working.



NEVER DO!



NEVER DO!

You are encouraged to tell us about new circuits you create. If they are unique, we will post them with your name and state on our website at www.elenco.com/showcase.

Send your suggestions (with photos) to info@elenco.com.

Elenco® provides a circuit designer so that you can make your own Snap Circuits® drawings. This Microsoft® Word document can be downloaded from www.elenco.com/for-makers.



WARNING: SHOCK HAZARD - Never connect Snap Circuits® to the electrical outlets in your home in any way!

Basic Troubleshooting

1. Most circuit problems are due to incorrect assembly, always double-check that your circuit exactly matches the drawing for it.
2. Be sure that parts with positive/negative markings are positioned as per the drawing.
3. Be sure that all connections are securely snapped.
4. Try replacing the batteries.

IMPORTANT:

If any parts are missing or damaged,
DO NOT RETURN TO RETAILER.
Go to elenco.com/replacement-parts or
e-mail us at support@elenco.com.

Note: A complete parts list is on
page 2 in this manual.

If you have any problems, contact:

ELENCO® ELECTRONICS, LLC

150 Carpenter Ave. Wheeling, IL 60090
(800) 533-2441

e-mail: support@elenco.com | elenco.com

**Elenco® is not responsible for
parts damaged due to incorrect wiring.**

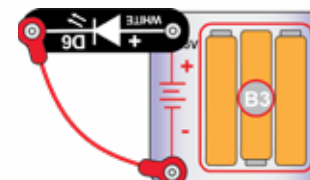
Advanced Troubleshooting

If you suspect you have damaged parts, you can follow this procedure to systematically determine which ones need replacing:

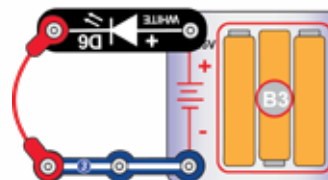
1. To Test battery holder (B3), white LED (D6) and Speaker (SP):

- Place batteries in holder.
- Place the **white LED** directly across the battery holder (LED "+" to battery "+"), it should light.
- "Tap" the **Speaker** across the battery holder contacts, you should hear static as it touches.
- If none work, then replace your batteries and repeat, if still bad then the **Battery Holder** is damaged

- ### 2. Jumper wires:
- Use this mini-circuit to test each jumper wire, the LED should light.



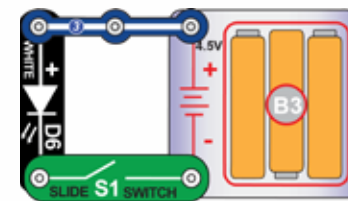
- ### 3. Snap wires:
- Use this mini-circuit to test each of the snap wires, one at a time. The LED should light.



4. Slide switch (S1), press switch (S2), vibration switch (S4), phototransistor (Q4), microphone (X1), and 5.1kΩ resistor (R3):

Use this mini-circuit; turning on S1 should light the white LED (D6) or S1 is bad.

- Replace S1 with press switch S2, pressing S2 should light the LED.
- Replace S2 with vibration switch S4, tapping on S4 should light the LED.
- Replace S4 with resistor R3, the LED should light but not be very bright.
- Replace R3 with phototransistor Q4 ("+" on left), vary the amount of light shing on Q4 to make the LED brighter or dimmer.
- Replace Q4 with X1 ("+" on left), the LED brightness should be similar.



5. **NPN transistor (Q2) and SCR (Q3):** Use project 7 to test them.
6. **Alarm IC (U2):** Use parts A & B of project 6 to test it. Each arrangement should produce a sound, or U2 is broken.
7. **Motion detector (U7):** Use project 1 to test it.
8. **Voice Changer (U34):** Use project 2 to test it.
9. **470μF capacitor (C5):** Touch C5 across the battery holder (B3) snaps (C5 "+" to B3 "+"), then touch C5 across the white LED (D6) snaps (C5 "+" to D6 "+"); the LED should light for a moment or C5 is broken.



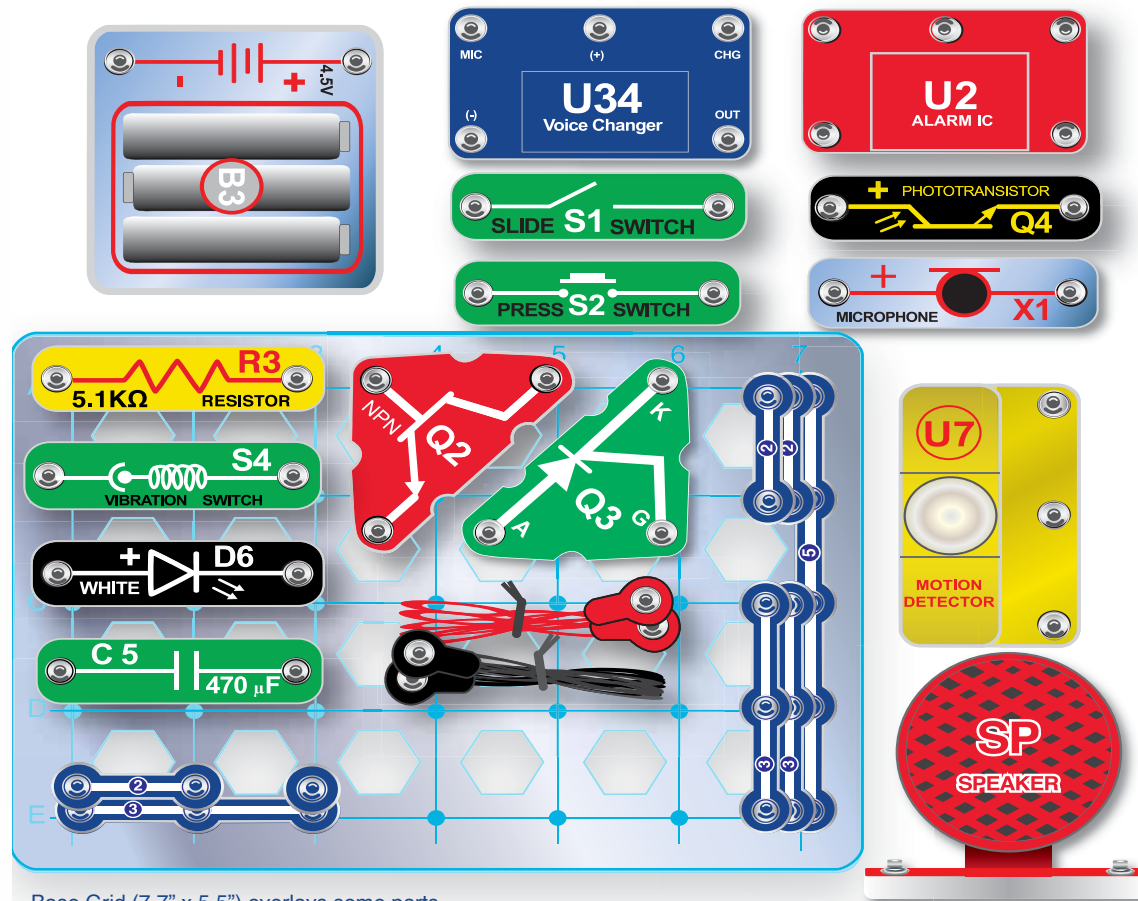
**ADULT
SUPERVISION
RECOMMENDED**

Snap Circuits® Spy Kit Parts Layout

IMPORTANT:

If any parts are missing or damaged, DO NOT RETURN TO RETAILER. Go to elenco.com/replacement-parts or e-mail us at support@elenco.com.

Note: A complete parts list is on page 2 in this manual.



Base Grid (7.7" x 5.5") overlays some parts.

If you have any problems, contact: ELENCO® ELECTRONICS, LLC
150 Carpenter Ave. Wheeling, IL 60090 | (800) 533-2441 | e-mail: support@elenco.com | elenco.com

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Conforms to all applicable U.S. government requirements and CAN ICES-3 (B)/NMB-3 (B).
Model: SCSPY1 | Part # 753083

